

1) Free electrons:

$$H_e = \sum_k \epsilon_k a_k^\dagger a_k, \quad \epsilon_k = \frac{\hbar^2 k^2}{2m}$$

$$H_p = \sum_{\lambda} \omega_{\lambda} \left(b_{\lambda}^\dagger b_{\lambda} + \frac{1}{2} \right)$$

$$N_{\lambda} = \langle b_{\lambda}^\dagger b_{\lambda} \rangle = \frac{1}{e^{\omega_{\lambda}/T} - 1}$$

$$H_{ei} = \sum_{\lambda} \int d\vec{r} \psi_i^\dagger(\vec{r}) \underbrace{U(\vec{r} - \vec{R}_\lambda)}_{U(\vec{r} - \vec{R}_\lambda^0) - \vec{A}_\lambda \cdot \vec{\nabla} U(\vec{r} - \vec{R}_\lambda^0)} \psi_i(\vec{r})$$

$$H_{e-i} = - \sum_{\lambda} \int d\vec{r} \psi_i^\dagger(\vec{r}) \left(\vec{A}_\lambda \cdot \vec{\nabla} U(\vec{r} - \vec{R}_\lambda^0) \right) \psi_i(\vec{r})$$

$$\vec{A}_\lambda = \frac{1}{\sqrt{N_i}} \sum_{\vec{z}\lambda} \sqrt{\frac{1}{2M\omega_{\lambda}}} \vec{E}_{\vec{z}\lambda} (b_{\vec{z}\lambda} + b_{-\vec{z}\lambda}^\dagger) e^{i\vec{z} \cdot \vec{R}_\lambda^0}$$

$$U = -\frac{ze^2}{r} e^{-\Lambda r}$$

$$\sum_{\vec{z}} e^{i(\vec{k} - \vec{k}' + \vec{z}) \cdot \vec{R}_\lambda^0} = N_i \sum_{\vec{G}} \delta_{\vec{k}, \vec{k}' + \vec{z} + \vec{G}}$$

$$- \sum_{\vec{k}, \vec{k}'} \sum_{\vec{z}\lambda} \frac{1}{\sqrt{2M\omega_{\lambda} N_i}} \vec{E}_{\vec{z}\lambda} \frac{c_{\vec{k}}^\dagger c_{\vec{k}'}^\dagger}{V} (b_{\vec{z}\lambda} + b_{-\vec{z}\lambda}^\dagger) e^{i\vec{z} \cdot \vec{R}_\lambda^0} \int d\vec{r} e^{i(\vec{k} - \vec{k}') \cdot \vec{r}} \underbrace{\vec{\nabla} \left(\frac{1}{|\vec{r} - \vec{R}_\lambda^0|} e^{-\Lambda |\vec{r} - \vec{R}_\lambda^0|} \right)}_{\substack{\vec{r} - \vec{R}_\lambda^0 = \vec{x} \\ \Lambda \vec{r} = -i\vec{x}}}$$

$$\leftarrow \sum_{\vec{z}} \vec{z} + \vec{G}$$

$$N_i = \frac{V}{\Omega}$$

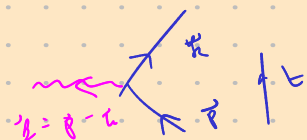
$$\sum_{\vec{k}, \vec{k}'} \sum_{\vec{z}\lambda} \frac{i \left[\vec{E}_{\vec{z}\lambda} \cdot (\vec{k} - \vec{k}') \right]}{\sqrt{2M\omega_{\lambda}}} \frac{1}{V} \sqrt{N_i} U(\vec{k} - \vec{k}') \sum_{\vec{G}} \delta_{\vec{k}, \vec{k}' + \vec{z} + \vec{G}} c_{\vec{k}}^\dagger c_{\vec{k}'}^\dagger (b_{\vec{z}\lambda} + b_{-\vec{z}\lambda}^\dagger) =$$

$$= \frac{1}{\sqrt{V}} \sum_{\vec{k}, \vec{z}\lambda} g_{\lambda}(\vec{k}, \vec{z}) c_{\vec{k}+\vec{z}}^\dagger c_{\vec{k}} (b_{\vec{z}\lambda} + b_{-\vec{z}\lambda}^\dagger) ; \quad g_{\lambda}(\vec{k}, \vec{z}) = -i \frac{1}{\sqrt{2M\omega_{\lambda}}} \frac{4\pi ze^2}{(|\vec{z}|^2 + \Lambda^2)} \vec{z} \cdot \vec{E}_{\vec{z}\lambda}$$

→ Fermi: spin ne drinja!

$$|\langle f | c_{\vec{k}}^\dagger c_{\vec{r}} | i \rangle|^2 = n_{\vec{r}} (1 - n_{\vec{k}})$$

polu prazno



→ from trace: $K N_z - 1 |b_z| N_z \rangle|^2 = N_z$
 $| \langle N_z | b_z^\dagger | N_z + 1 \rangle |^2 = N_z + 1$

Fermi (variance): $\gamma_{i \rightarrow f} = 2\pi |K f | H_{fi} | i \rangle|^2 \underbrace{p(E_f)}_{\delta(E_f - E_i)}$

(1) Absorption process:

$|i\rangle = |N_z\rangle \otimes |k\rangle$
 $|f\rangle = |N_z+1\rangle \otimes |k+\vec{z}\rangle$
 $\rightarrow \sum_{k_z} \underbrace{\langle N_z+1 | (b_z + b_{-z}^\dagger) | N_z \rangle}_{N_z} \underbrace{\langle k+\vec{z} | c_{k+\vec{z}}^\dagger c_k | k \rangle}_{\sqrt{n_k (1-n_{k+\vec{z}})}} \cdot g \frac{1}{\sqrt{V}}$

(2) emission process:

$|i\rangle = |N_z\rangle |k\rangle$
 $|f\rangle = |N_z+1\rangle |k-\vec{z}\rangle$
 $\rightarrow \sum_{k_z} \underbrace{\langle N_z+1 | b_z^\dagger | N_z \rangle}_{\sqrt{N_z+1}} \underbrace{\langle k-\vec{z} | c_{k-\vec{z}}^\dagger c_k | k \rangle}_{\sqrt{n_k (1-n_{k-\vec{z}})}} \cdot g \frac{1}{\sqrt{V}}$

(Mixture? : $c_{k+\vec{z}}^\dagger c_k b_z + c_{k-\vec{z}}^\dagger c_k b_z^\dagger$)

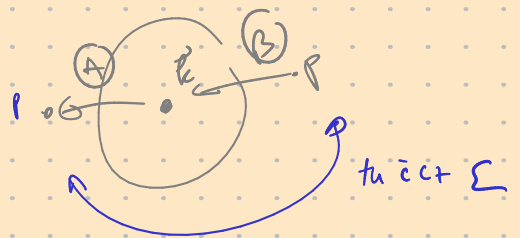
(A) $k \rightarrow p$: $\gamma_{k \rightarrow p} = \gamma_{abs} + \gamma_{em} = \frac{2\pi}{V} |g_\lambda(\vec{k}, \vec{z})|^2 n_k (1-n_{k+\vec{z}}) N_z \delta(\epsilon_{k+\vec{z}} - \epsilon_k - \omega_z) + \frac{2\pi}{V} |g_\lambda(\vec{k}, \vec{z})|^2 n_k (1-n_{k-\vec{z}}) (N_z+1) \delta(\epsilon_{k-\vec{z}} + \omega_z - \epsilon_k)$

(B) $p \rightarrow k$: $\gamma_{p \rightarrow k} = \gamma_{abs} + \gamma_{em} = \frac{2\pi}{V} |g_\lambda(\vec{k}, \vec{z})|^2 n_{k+\vec{z}} (1-n_k) (N_z+1) \delta(\epsilon_k + \omega_z - \epsilon_{k+\vec{z}}) + \frac{2\pi}{V} |g_\lambda(\vec{k}, \vec{z})|^2 n_{k-\vec{z}} (1-n_k) N_z \delta(\epsilon_k - \epsilon_{k-\vec{z}} - \omega_z)$

$N_{-z} = N_z$ then $\omega_z = \omega_{-z}$?

power inv. strength

$I_k = \sum_p \gamma_{p \rightarrow k} - \gamma_{k \rightarrow p} = \dots$



$= \frac{2\pi}{V} \sum_{z, \lambda} |g_\lambda(\vec{k}, \vec{z})|^2 \left\{ (1-n_k) n_{k+\vec{z}} (1+N_z) \delta(\epsilon_k - \epsilon_{k+\vec{z}} + \omega_z) + (1-n_k) n_{k-\vec{z}} N_z \delta(\epsilon_k - \epsilon_{k-\vec{z}} - \omega_z) - n_k (1-n_{k-\vec{z}}) (1+N_z) \delta(\epsilon_{k-\vec{z}} - \epsilon_k + \omega_z) - n_k (1-n_{k+\vec{z}}) N_z \delta(\epsilon_{k+\vec{z}} - \epsilon_k - \omega_z) \right\}$

① $q \rightarrow -q$ $\hat{n}_k \rightarrow \hat{n}_{-k}$ $\hat{n}_k \rightarrow \hat{n}_{-k}$ $\hat{n}_k \rightarrow \hat{n}_{-k}$ $\delta(x-y) = \delta(y-x)$

$$\dot{n}_k = \frac{2\pi}{V} \sum_{k'} |g_\lambda(k, k')|^2 \left\{ (1-n_k) n_{k'} (1+N_2) \delta(\epsilon_k - \epsilon_{k'} + \omega_2) + \right. \\ \left. + (1-n_k) n_{k'} N_2 \delta(\epsilon_k - \epsilon_{k'} - \omega_2) - \right. \\ \left. - n_k (1-n_{k'}) (1+N_2) \delta(\epsilon_k + \epsilon_{k'} - \omega_2) - \right. \\ \left. - n_k (1-n_{k'}) N_2 \delta(\epsilon_k + \epsilon_{k'} + \omega_2) \right\} =$$

$$= \frac{2\pi}{V} \sum_{k'} |g_\lambda(k, k')|^2 \left\{ \delta(\epsilon_k - \epsilon_{k'} - \omega_2) \left[(1-n_k) n_{k'} N_2 - (1-n_{k'}) n_k (1+N_2) \right] + \right. \\ \left. + \delta(\epsilon_k - \epsilon_{k'} + \omega_2) \left[(1-n_k) n_{k'} (1+N_2) - n_k (1-n_{k'}) N_2 \right] \right\} =$$

Bei demselben:

✓ analog hier (1.23b)

$$\dot{n}_k = \sum_{k'} \Gamma_{k' \rightarrow k}^{\lambda} (1-n_k) n_{k'} - \Gamma_{k \rightarrow k'}^{\lambda} (1-n_{k'}) n_k$$

$$\Gamma_{k' \rightarrow k}^{\lambda} = \frac{2\pi}{V} |g_\lambda(k, k')|^2 \left\{ (1+N_2) \delta(\epsilon_k - \epsilon_{k'} + \omega_2) + N_2 \delta(\epsilon_k - \epsilon_{k'} - \omega_2) \right\}$$

$$\Gamma_{k \rightarrow k'}^{\lambda} = \frac{2\pi}{V} |g_\lambda(k, k')|^2 \left\{ (1+N_2) \delta(\epsilon_{k'} - \epsilon_k + \omega_2) + N_2 \delta(\epsilon_{k'} - \epsilon_k - \omega_2) \right\}$$

2. način:

$$n_k = \text{Tr}[\rho_e \cdot c_k^\dagger c_k]$$

$$N_z = \text{Tr}[\rho_e \cdot b_z^\dagger b_z]$$

$$\rho_0 = \rho_e \otimes \rho_h$$

$$H_{eq} = \frac{1}{\sqrt{V}} \sum_{k,q,\lambda} g_\lambda(k,q) c_{k+q}^\dagger c_k (b_{2\lambda} + b_{-2\lambda}^\dagger) \quad ; \quad \rho \equiv k_1 \rho$$

$$c_k(t) = e^{-i\varepsilon_k t} c_k \rightarrow c_q^\dagger c_k(t) = e^{i(\varepsilon_q - \varepsilon_k)t} c_q^\dagger c_k$$

$$b_{2\lambda}(t) = e^{-i\omega_{2\lambda} t} b_{2\lambda}$$

$$\rightarrow H_{eq}^I(t) = \frac{1}{\sqrt{V}} \sum_{k,q,\lambda} g_\lambda(k,q) c_q^\dagger c_k (b_{2\lambda} e^{i(\varepsilon_q - \varepsilon_k - \omega_{2\lambda})t} + b_{-2\lambda}^\dagger e^{i(\varepsilon_q - \varepsilon_k + \omega_{2\lambda})t})$$

$$\rightarrow I_k^I(t) = \dot{n}_k^I(t) = i [H_{eq}^I(t), n_k] \quad \text{in} \quad \text{Tr}[\rho_0 I_k^I] = 0, \quad \text{in} \quad I \propto \langle b_{2\lambda} + b_{-2\lambda}^\dagger \rangle = 0$$

- $\mathcal{O}(\omega^2)$:

$$\dot{n}_k = i^2 \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T dt \text{Tr}(\rho_0 [H_{eq}^I(t), [H_{eq}, n_k]])$$

$[c_k, n_k] = \delta_{kk} c_k$
 $[c_k^\dagger, n_k] = -\delta_{kk} c_k^\dagger$

$$(1) \quad \frac{1}{\sqrt{V}} \sum_{k,q,\lambda} g_\lambda(k,q) (b_{2\lambda} + b_{-2\lambda}^\dagger) \underbrace{[c_{k+q}^\dagger c_k, n_k]}_{c_{k+q}^\dagger c_k \delta_{kk} - c_k^\dagger c_{k+q} \delta_{k+q,k}} =$$

$$= \frac{1}{\sqrt{V}} \sum_{q,\lambda} g_\lambda^{(1)}(q) (b_{2\lambda} + b_{-2\lambda}^\dagger) (c_{k+q}^\dagger c_k - c_k^\dagger c_{k+q})$$

$$(2) \quad \frac{1}{V} \sum_{k',q',\lambda'} \sum_{k,\lambda} g_{\lambda'}(k',q') g_\lambda(k,q) \left\{ (b_{2\lambda'}(t) + b_{-2\lambda'}^\dagger(t)) (b_{2\lambda} + b_{-2\lambda}^\dagger) [c_{k'+q'}^\dagger c_{k'}, c_{k+q}^\dagger c_k - c_k^\dagger c_{k+q}] + \right.$$

$$\left. + [b_{2\lambda'}(t) + b_{-2\lambda'}^\dagger(t), b_{2\lambda} + b_{-2\lambda}^\dagger] \cdot c_{k'+q'}^\dagger c_{k'} (c_{k+q}^\dagger c_k - c_k^\dagger c_{k+q}) \right\}$$

... born karnije, pravi deo.

2. BCS currents za ph-el. coupling:

$$k = k_{\uparrow\downarrow} \quad -k = k_{\downarrow\uparrow}$$

$$H_e = \sum_k \xi_k (c_k^\dagger c_k + c_{-k}^\dagger c_{-k}) - \sum_k \left(\Delta c_k^\dagger c_{-k}^\dagger + \Delta^* c_k c_{-k} \right) \quad \begin{matrix} \xi_k = \epsilon_k - \mu \\ \xi_{-k} = \xi_k \end{matrix}$$

$$\rightarrow E_k = \sqrt{\xi_k^2 + \Delta^2}, \quad M_k = \frac{1}{2} \left(1 + \frac{\xi_k}{E_k} \right), \quad N_k = \frac{1}{2} \left(1 - \frac{\xi_k}{E_k} \right), \quad M_k N_k = \frac{\Delta}{2E_k}$$

$$\rightarrow \begin{pmatrix} c_k \\ c_{-k}^\dagger \end{pmatrix} = \begin{pmatrix} M_k & N_k \\ -N_k & M_k \end{pmatrix} \begin{pmatrix} \gamma_k \\ \gamma_{-k}^\dagger \end{pmatrix} \Rightarrow H_e = E_0 + \sum_{k\sigma} E_k \gamma_{k\sigma}^\dagger \gamma_{k\sigma} \quad N_{k\sigma} = \langle \gamma_{k\sigma}^\dagger \gamma_{k\sigma} \rangle$$

→ enaka interakcija: $H_{ep} = \frac{1}{V} \sum_{\vec{k}, \vec{k}', \sigma} g_{\lambda}(\vec{k}, \vec{k}') c_{\vec{k}+\vec{k}', \sigma}^\dagger c_{\vec{k}, \sigma} (b_{\vec{k}'}^\dagger + b_{-\vec{k}'}^\dagger)$

rotiramo

rotiramo s novo bazo: $(M_k' = M_{k+2}, N_k' = N_{k+2}) \quad (M_k' = M_{-k}, N_k' = N_{-k}!)$

$$\begin{aligned} c_{k\uparrow} &= M_k \gamma_{k\uparrow} + N_k \gamma_{-k\downarrow}^\dagger \\ c_{k\uparrow}^\dagger &= M_k \gamma_{k\uparrow}^\dagger + N_k \gamma_{-k\downarrow}^\dagger \end{aligned}$$

$$\begin{aligned} c_{k\downarrow} &= M_k \gamma_{k\downarrow} - N_k \gamma_{-k\uparrow}^\dagger \\ c_{k\downarrow}^\dagger &= M_k \gamma_{k\downarrow}^\dagger - N_k \gamma_{-k\uparrow}^\dagger \end{aligned}$$

$$\begin{aligned} \downarrow \\ \sum_{k\sigma} c_{k+\vec{q}, \sigma}^\dagger c_{k\sigma} &= \sum_k \left[(M_{k+2} \gamma_{k+2\uparrow}^\dagger + N_{k+2} \gamma_{-k-2\downarrow}^\dagger) (M_k \gamma_{k\uparrow} + N_k \gamma_{-k\downarrow}^\dagger) + (M_{k+2} \gamma_{k+2\downarrow}^\dagger - N_{k+2} \gamma_{-k-2\uparrow}^\dagger) (M_k \gamma_{k\downarrow} - N_k \gamma_{-k\uparrow}^\dagger) \right] \\ &= \sum_{k\sigma} M_{k+2} M_k (\gamma_{k+2\uparrow}^\dagger \gamma_{k\sigma}) + \sum_{k\sigma} N_{k+2} N_k (\gamma_{-k-2\downarrow}^\dagger \gamma_{-k\sigma}^\dagger) + \\ &\quad + \sum_k (M_{k+2} N_k \gamma_{k+2\uparrow}^\dagger \gamma_{-k\downarrow}^\dagger - N_{k+2} M_k \gamma_{k+2\downarrow}^\dagger \gamma_{-k\uparrow}^\dagger) \stackrel{h \rightarrow -q-h}{=} (M_{k+2} N_k + M_k N_{k+2}) \gamma_{k+2\uparrow}^\dagger \gamma_{-k\downarrow}^\dagger \\ &\quad + \sum_k (N_{k+2} M_k \gamma_{-k-2\downarrow}^\dagger \gamma_{k\uparrow} - M_{k+2} N_k \gamma_{-k-2\uparrow}^\dagger \gamma_{k\downarrow}) \stackrel{h \rightarrow -q-h}{=} (M_{k+2} N_k + M_k N_{k+2}) \gamma_{-k-2\downarrow}^\dagger \gamma_{k\uparrow} \end{aligned}$$

$$\begin{aligned} P_2^+ &= \sum_{k\sigma} c_{k\sigma}^\dagger c_{k\sigma} = 2 \delta_{\vec{q}, 0} \sum_k N_k^2 + \sum_{k\sigma} (M_{k+2} M_k - N_{k+2} N_k) \gamma_{k+2, \sigma}^\dagger \gamma_{k, \sigma} + \\ &\quad + \sum_k (M_{k+2} N_k + N_{k+2} M_k) [\gamma_{k+2\uparrow}^\dagger \gamma_{-k\downarrow}^\dagger + \gamma_{-k-2\downarrow}^\dagger \gamma_{k\uparrow}] \end{aligned}$$

scattering
ce ali ne

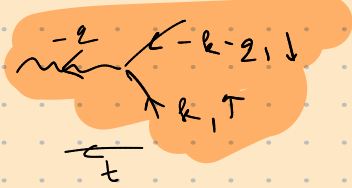
• scattering člena enaka, o faktorju: $|M_{k+2} M_k - N_{k+2} N_k|^2$

$$\hookrightarrow g_{\lambda}(\vec{k}, \vec{k}') \rightarrow G_{\lambda}^s(\vec{k}, \vec{k}') = g_{\lambda} (M_{k+2} M_k - N_{k+2} N_k)$$

• pair creation: $H_{eg}^c = \frac{1}{V} \sum_{k,q,\lambda} \left[\underbrace{G_{\lambda}^c(k,q)}_{G_{\lambda}^c(k,q)} (n_{k+q,\uparrow} n_{k,\downarrow} + n_{k+q,\downarrow} n_{k,\uparrow}) \right] \gamma_{k+q,\uparrow}^{\dagger} \gamma_{k,\downarrow}^{\dagger} (b_{2\lambda} + b_{-2\lambda}^{\dagger})$

$|\langle f | H_{eg}^c | i \rangle|^2 \propto \underbrace{|N_{2-1} \langle b_{2\lambda} | N_{2-} \rangle|^2}_{N_2} \cdot (1 - n_{k+q,\uparrow}) (1 - n_{k,\downarrow})$

• pair annihilation: $H_{eg}^a \propto \gamma_{k+q,\downarrow} \gamma_{k,\uparrow} (b_{2\lambda} + b_{-2\lambda}^{\dagger})$



$|\langle f | H_{eg}^a | i \rangle|^2 \propto \underbrace{|N_{2-1} \langle b_{2\lambda}^{\dagger} | N_{2-} \rangle|^2}_{(N_{2-}+1)} \cdot n_{k,\uparrow} n_{-k-q,\downarrow}$

calculate for:

$p = k + q$

$\dot{n}_k = \sum_p \Gamma_{p \rightarrow k}^s (1 - n_k) n_p - \Gamma_{k \rightarrow p}^s n_k (1 - n_p) + \Gamma_{k,p}^c (1 - n_{k+q,\uparrow}) (1 - n_{k,\downarrow}) - \Gamma_{k,p}^a n_{k,\uparrow} n_{-(k+q),\downarrow}$

$\Gamma_{p \rightarrow k}^s = \frac{2\pi}{V} \sum_{\lambda} |G_{\lambda}^s(k,p)|^2 \cdot \left\{ (1 + N_2) \delta(E_k - E_p + \omega_2) + N_{2-} \delta(E_k - E_p - \omega_2) \right\}$

$\Gamma_{k \rightarrow p}^s = \frac{2\pi}{V} \sum_{\lambda} |G_{\lambda}^s(k,p)|^2 \cdot \left\{ (1 + N_{2-}) \delta(E_p - E_k + \omega_2) + N_2 \delta(E_p - E_k - \omega_2) \right\}$

$\Gamma_{k,p}^c = \frac{2\pi}{V} \sum_{\lambda} |G_{\lambda}^c(k,p)|^2 \cdot \left\{ N_2 \delta(E_p + E_k - \omega_2) \right\}$

$\Gamma_{k,p}^a = \frac{2\pi}{V} \sum_{\lambda} |G_{\lambda}^a(k,p)|^2 \cdot \left\{ (1 + N_{2-}) \delta(\omega_2 - E_p - E_k) \right\}$

→ v. standard major: $(u, v) \rightarrow (u, v)$ in (u, v) (major: her pr. n. j. h. : $\begin{pmatrix} u & -v \\ v & u \end{pmatrix}$ pr. m. m. : $\begin{pmatrix} u & v \\ -v & u \end{pmatrix}$).

→ key je: $A_{\gamma-\delta}$ (namer $\delta(E_p - E_{p'} - \omega_2) \cdot \delta(p - p' - \frac{\omega_2}{c})$???

3. EI currents, el. - ph.

$$H_e = \sum_k (c_{a,k}^\dagger, c_{b,k}^\dagger) \begin{pmatrix} \epsilon_a + v n_b & -\Delta^\dagger \\ -\Delta & \epsilon_b + v n_a \end{pmatrix} \begin{pmatrix} c_{a,k} \\ c_{b,k} \end{pmatrix}$$

$$\hookrightarrow \begin{pmatrix} c_{a,k} \\ c_{b,k} \end{pmatrix} = \begin{pmatrix} m_k & r_k \\ -v_k & m_k \end{pmatrix} \begin{pmatrix} \alpha_k \\ \beta_k \end{pmatrix}, \quad E_{\lambda k} = v_k \pm \sqrt{\xi_k^2 + \Delta^2};$$

$$\eta_k = \frac{\epsilon_a + \epsilon_b + v(n_a + n_b)}{2}, \quad \xi_k = \frac{\epsilon_a - \epsilon_b + v(n_a - n_b)}{2}$$

?

$$H_{ep} = \frac{1}{\sqrt{V}} \sum_{k, \lambda, \lambda'} (c_{a,k+\lambda}^\dagger, c_{b,k+\lambda}^\dagger) \begin{pmatrix} g_{a,\lambda}(k, \lambda) & 0 \\ 0 & g_{b,\lambda}(k, \lambda) \end{pmatrix} \begin{pmatrix} c_{a,k} \\ c_{b,k} \end{pmatrix} (b_{\lambda,1}^\dagger + b_{-\lambda,1}^\dagger)$$

$$\begin{aligned} c_{a,k+\lambda}^\dagger c_{a,k} &= (u_p \alpha_k^\dagger + v_p \beta_k^\dagger) (u_k \alpha_k + v_k \beta_k) = \\ &= u_p u_k \alpha_k^\dagger \alpha_k + v_p v_k \beta_k^\dagger \beta_k + u_p v_k \alpha_k^\dagger \beta_k + v_p u_k \beta_k^\dagger \alpha_k \end{aligned}$$

$$H_{ep} = \frac{1}{\sqrt{V}} \sum_{k, \lambda, \lambda'} (\alpha_{k+\lambda}^\dagger, \beta_{k+\lambda}^\dagger) \begin{pmatrix} g_{a,\lambda}(k, \lambda) u_k u_{k+\lambda} + g_{b,\lambda}(k, \lambda) v_k v_{k+\lambda} & g_{a,\lambda}(k, \lambda) v_k v_{k+\lambda} - g_{b,\lambda}(k, \lambda) u_k u_{k+\lambda} \\ g_{a,\lambda}(k, \lambda) v_k v_{k+\lambda} - g_{b,\lambda}(k, \lambda) u_k u_{k+\lambda} & g_{a,\lambda}(k, \lambda) u_k u_{k+\lambda} + g_{b,\lambda}(k, \lambda) v_k v_{k+\lambda} \end{pmatrix} \begin{pmatrix} \alpha_k \\ \beta_k \end{pmatrix} (b_{\lambda,1}^\dagger + b_{-\lambda,1}^\dagger)$$

$$G_{\mu\nu}(k, \lambda) = U_{\mu\lambda} (g_{\mu\nu} \delta_{\lambda\lambda'}) U_{\lambda'\nu} \quad \begin{matrix} \lambda, \omega = a, b \\ \lambda, \mu = a, b \end{matrix}$$

→ Fermi: $\mu, \nu = \{a, b\}$

$$\Gamma_{\mu k \rightarrow \nu k} = 2\pi \sum_i |\langle f | H_{ep} | i \rangle|^2 \delta(E_f - E_i)$$

$$G_{\mu\nu} \hat{p}_{\mu, k+\lambda}^\dagger \hat{p}_{\nu, k} = \text{diagram} \quad \frac{1}{V} |G_{\mu\nu}|^2 n_{\mu k} (1 - n_{\mu, k+\lambda}) N_{\lambda, 2}$$

$$G_{\mu\nu} \hat{p}_{\mu, k+\lambda}^\dagger \hat{p}_{\nu, k} = \text{diagram} \quad \frac{1}{V} |G_{\mu\nu}|^2 n_{\nu k} (1 - n_{\mu, k+\lambda}) \cdot (1 + N_{\lambda, 2})$$

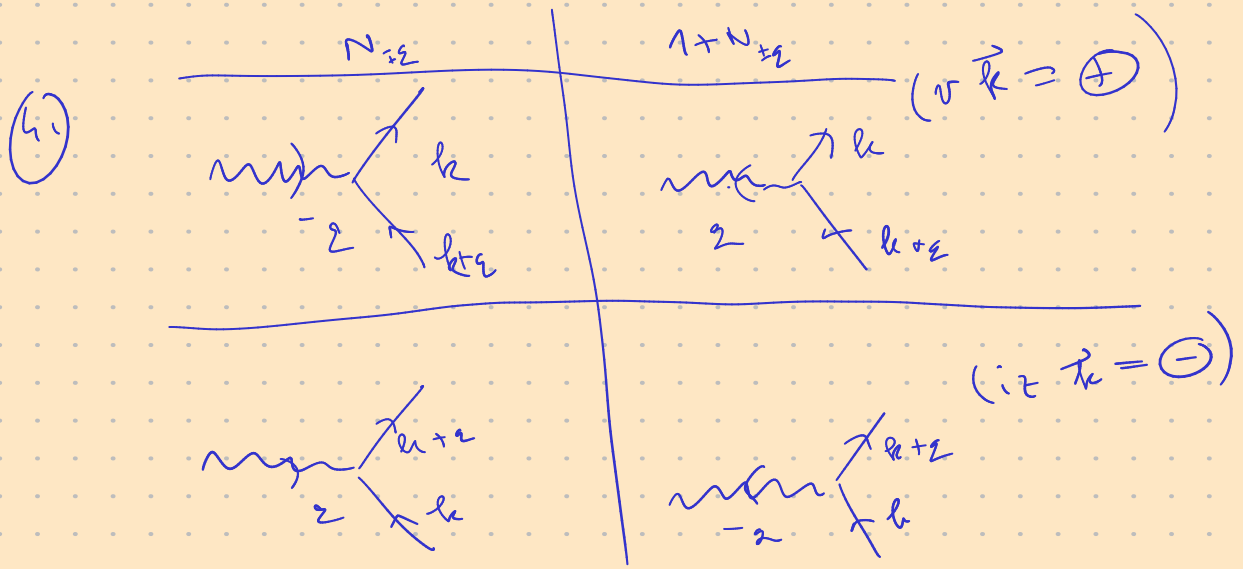
$\Gamma_{\nu p \rightarrow \mu k} = \text{diagram}$

$k \rightarrow (k) \xrightarrow{p}$

$$\dot{n}_{ak} = \sum_p \left(\Gamma_{ap \rightarrow ak} n_{ap} (1 - n_{ak}) - \Gamma_{ak \rightarrow ap} n_{ak} (1 - n_{ap}) + \Gamma_{bp \rightarrow ak} n_{bp} (1 - n_{ak}) - \Gamma_{ak \rightarrow bp} n_{ak} (1 - n_{bp}) \right)$$

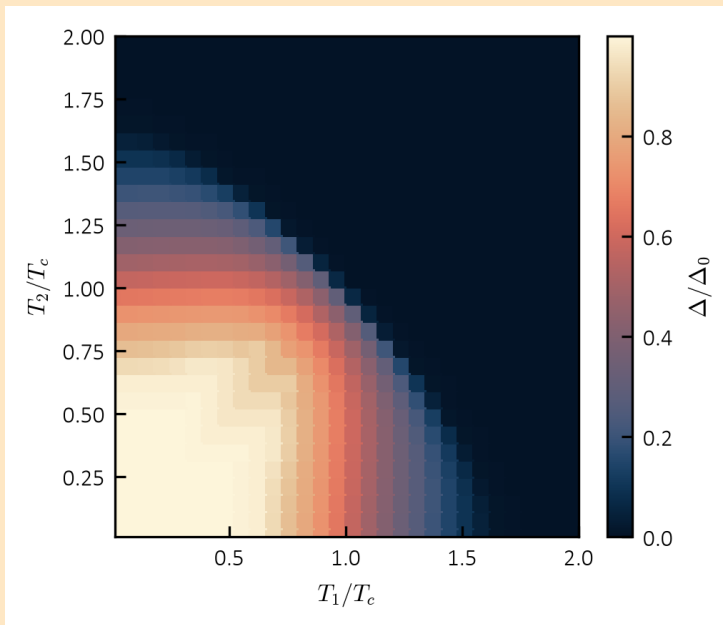
$$\Gamma_{\mu p \rightarrow \nu k} = \frac{2\pi}{V} \sum_{\lambda} |G_{\mu\nu}^\lambda(k, \lambda)|^2 \left\{ (1 + N_{\lambda, 2}) \delta(E_{\nu k} - E_{\mu p} + \omega_{\lambda, 2}) + N_{\lambda, 2} \delta(E_{\nu k} - E_{\mu p} - \omega_{\lambda, 2}) \right\}$$

$$\Gamma_{\mu k \rightarrow \nu p} = \frac{2\pi}{V} \sum_{\lambda} |G_{\mu\nu}^\lambda(k, \lambda)|^2 \left\{ (1 + N_{\lambda, 2}) \delta(E_{\mu p} - E_{\nu k} + \omega_{\lambda, 2}) + N_{\lambda, 2} \delta(E_{\mu p} - E_{\nu k} - \omega_{\lambda, 2}) \right\}$$



$$G^{\Lambda}(\vec{k}, \vec{z}) = \begin{pmatrix} \mu_{k+z} & \nu_{k+z} \\ -\nu_{k+z} & \mu_{k+z} \end{pmatrix} \begin{pmatrix} g_{\Lambda}(\vec{z}) & 0 \\ 0 & g_{\Lambda}(\vec{z}) \end{pmatrix} \begin{pmatrix} \mu_k & \nu_k \\ -\nu_k & \mu_k \end{pmatrix}$$

$$g_{\Lambda}(\vec{z}) = -i \frac{\hbar \omega_c e^2}{\sqrt{2M\omega_{\Lambda}}} \cdot \frac{1}{|\vec{z}|^2 + \Lambda^2} (\vec{z} \cdot \vec{e}_{2\Lambda}^*)$$



4. free el. - photon int.

($e = -e$)

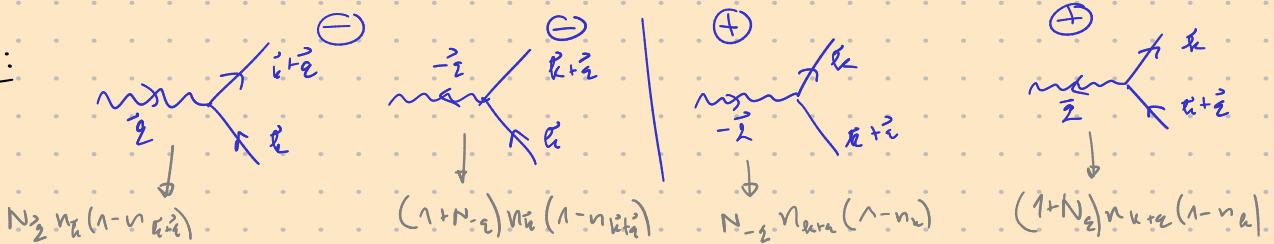
$$H = \frac{1}{2m} (\vec{p} + e\vec{A})^2 = \frac{p^2}{2m} + \underbrace{\frac{e}{2m} \vec{p} \cdot \vec{A}}_{\substack{\text{"} \\ -(\vec{p} \cdot \vec{A}) - i\vec{A} \cdot \vec{p} = \vec{A} \cdot \vec{p}}} + \frac{e}{2m} \vec{A} \cdot \vec{p} + \frac{e^2}{2m} A^2 = \frac{p^2}{2m} + \frac{e}{m} (\vec{A} \cdot \vec{p}) + \underbrace{\frac{e^2}{2m} A^2}_{\approx 0} \quad \mathcal{O}(A^2)$$

$$H = \sum_{\vec{k}\sigma} \epsilon_{\vec{k}} c_{\vec{k}\sigma}^\dagger c_{\vec{k}\sigma} + \frac{e}{m} \sum_{\vec{k}\sigma} \int d\vec{r} \psi_\sigma^\dagger(\vec{r}) [\vec{A} \cdot \vec{p}] \psi_\sigma(\vec{r})$$

$$\sum_{\vec{k}\sigma} \sum_{\vec{k}'\sigma'} \frac{e}{m} \int d\vec{r} e^{i(\vec{k}-\vec{k}')\cdot\vec{r}} c_{\vec{k}'\sigma'}^\dagger \left[\frac{1}{2\epsilon_0 V \omega_{\vec{k}'}} \vec{\epsilon}_{\vec{k}'} (a_{\vec{k}} e^{i\vec{k}\cdot\vec{r}} + a_{-\vec{k}}^\dagger e^{i\vec{k}\cdot\vec{r}}) \right] c_{\vec{k}\sigma} =$$

$$= \frac{1}{V} \sum_{\vec{k}\sigma} \frac{e}{\sqrt{2\epsilon_0 \omega_{\vec{k}}}} (\vec{\epsilon}_{\vec{k}} \cdot (\vec{k} + \vec{k}')) \cdot c_{\vec{k}+\vec{k}'\sigma}^\dagger c_{\vec{k}\sigma} (a_{\vec{k}} + a_{-\vec{k}}^\dagger) = 11 e\gamma^{(1)}$$

→ Fermi:



$$\sum_{\vec{k}} T_{\vec{k} \rightarrow \vec{k}'} = \sum_{\vec{k}, \sigma} \left(\frac{e^2}{2\epsilon_0 m^2 V \omega_{\vec{k}}} |\vec{\epsilon}_{\vec{k}} \cdot \vec{k}|^2 \right) \left\{ N_{-\vec{k}} \delta(\epsilon_k - \epsilon_{\vec{k}'} - \omega_{\vec{k}}) + (1 + N_{\vec{k}}) \delta(\epsilon_k - \epsilon_{\vec{k}'} + \omega_{\vec{k}}) \right\}$$

$$\sum_{\vec{k}} T_{\vec{k} \rightarrow \vec{k}'} = \sum_{\vec{k}, \sigma} \left(\frac{e^2}{2\epsilon_0 m^2 V \omega_{\vec{k}}} |\vec{\epsilon}_{\vec{k}} \cdot \vec{k}|^2 \right) \left\{ N_{\vec{k}} \delta(\epsilon_{\vec{k}'} - \epsilon_k - \omega_{\vec{k}}) + (1 + N_{-\vec{k}}) \delta(\epsilon_{\vec{k}'} - \epsilon_k + \omega_{-\vec{k}}) \right\}$$

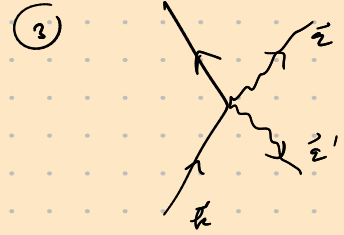
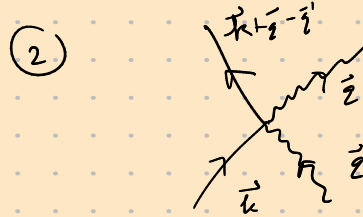
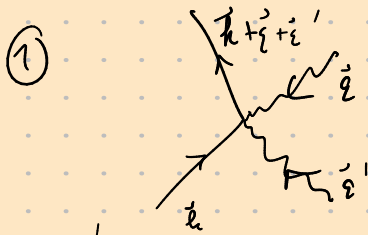
$$\mathcal{O}(A^2): H_{e\gamma}^{(2)} = \frac{e^2}{2m} \sum_{\vec{k}\sigma} \int d\vec{r} \psi_\sigma^\dagger(\vec{r}) (\vec{A}(\vec{r}) \cdot \vec{A}(\vec{r})) \psi_\sigma(\vec{r}) =$$

$$= \frac{e^2}{2m} \sum_{\vec{k}\sigma} \sum_{\vec{k}'\sigma'} \int d\vec{r} e^{i(\vec{k}-\vec{k}')\cdot\vec{r}} c_{\vec{k}'\sigma'}^\dagger \left[\sum_{\vec{k}''\sigma''} \frac{1}{2\epsilon_0 V} \frac{1}{\sqrt{\omega_{\vec{k}''} \omega_{\vec{k}'''}}} (\vec{\epsilon}_{\vec{k}''} \cdot \vec{\epsilon}_{\vec{k}'''}) \cdot (a_{\vec{k}''} e^{i\vec{k}''\cdot\vec{r}} + a_{-\vec{k}''}^\dagger e^{i\vec{k}''\cdot\vec{r}}) (a_{\vec{k}'''} e^{i\vec{k}'''\cdot\vec{r}} + a_{-\vec{k}'''}^\dagger e^{i\vec{k}'''\cdot\vec{r}}) \right] c_{\vec{k}\sigma}$$

$$\textcircled{1} \sum_{\substack{\vec{k}''\sigma'' \\ \vec{k}'''\sigma'''}} \int d\vec{r} e^{i(\vec{k}-\vec{k}'+\vec{k}''+\vec{k}''')\cdot\vec{r}} \text{ or } c_{\vec{k}'\sigma'}^\dagger c_{\vec{k}\sigma} a_{\vec{k}''} a_{\vec{k}'''}$$

$$(a_{\vec{k}} a_{\vec{k}'} + a_{\vec{k}} a_{-\vec{k}'}^\dagger + a_{-\vec{k}}^\dagger a_{\vec{k}'} + a_{-\vec{k}}^\dagger a_{-\vec{k}'}^\dagger) e^{i(\vec{k}+\vec{k}')\cdot\vec{r}} \sim \sum_{\vec{k}''\sigma''} \delta_{\vec{k}', \vec{k}+\vec{k}''} a_{\vec{k}} a_{\vec{k}''}$$

$$H_{\text{er}}^{(2)} = \frac{e^2}{4 \epsilon_0 V m} \sum_{\substack{\vec{k}, \vec{q}, \vec{q}' \\ \sigma, \sigma', \lambda, \lambda'}} \frac{\vec{\epsilon}_{\vec{q}, \lambda} \cdot \vec{\epsilon}_{\vec{q}', \lambda'}}{\sqrt{\omega_{\vec{q}} \omega_{\vec{q}'}}} \cdot \left[\overset{(1)}{c_{\vec{k}+\vec{q}+\vec{q}', \sigma}^+ c_{\vec{k}, \sigma} a_{\vec{q}, \lambda} a_{\vec{q}', \lambda'}}} + \overset{(2)}{2 a_{\vec{q}, \lambda}^+ a_{\vec{q}', \lambda'}} c_{\vec{k}+\vec{q}-\vec{q}', \sigma}^+ c_{\vec{k}, \sigma}} + \right. \\ \left. + c_{\vec{k}-\vec{q}-\vec{q}', \sigma}^+ c_{\vec{k}, \sigma} \overset{(3)}{a_{\vec{q}, \lambda}^+ a_{\vec{q}', \lambda'}}} \right]$$



→ Fermi: $\langle f | H_{\text{er}}^{(2)} | i \rangle^2$

(1) $\sqrt{n_k (1 - n_{k+q+q'})} N_{\vec{q}, \lambda} N_{\vec{q}', \lambda'}$

(2) $\sqrt{n_k (1 - n_{k+q-q'})} N_{\vec{q}} (N_{\vec{q}'} + 1)$

(3) $\sqrt{n_k (1 - n_{k-q-q'})} (1 + N_{\vec{q}}) (1 + N_{\vec{q}'})$

$$\dot{n}_{k\sigma} = \sum_{\substack{\vec{q}, \vec{q}' \\ \lambda, \lambda'}} \frac{2\pi}{V} \left(\frac{e^2}{4 m \epsilon_0 V} \right)^2 \frac{(\vec{\epsilon}_{\vec{q}, \lambda} \cdot \vec{\epsilon}_{\vec{q}', \lambda'})}{\omega_{\vec{q}} \omega_{\vec{q}'}} N(\omega_{\vec{q}}) N(\omega_{\vec{q}'}) \cdot \delta(\epsilon_k - \epsilon_{k+q+q'} - \omega_{\vec{q}} - \omega_{\vec{q}'}) n_{k+q+q'} (1 - n_k) +$$

$$+ - \dots 2(1 + N_{\vec{q}}) N_{\vec{q}'} \delta \dots$$

$$+ (1 + N_{\vec{q}})(1 + N_{\vec{q}'} - (k \leftrightarrow q))$$

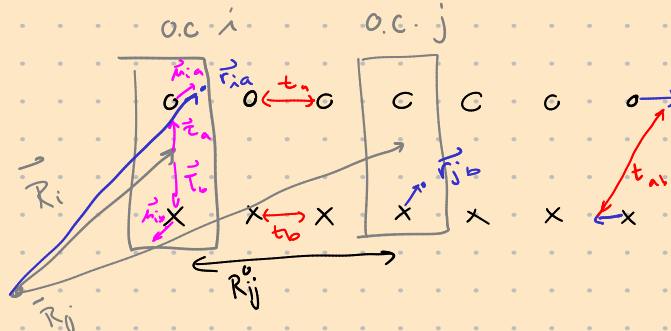
Interaction picture formalism:

$$H = H_0 + H'$$

$$\begin{aligned} \frac{dH_{\text{int}}}{dt} &= \text{Tr}(n_{\text{int}} \dot{p}_{\pm}) = (-i)^2 \int_0^t dt' \text{Tr} \left(n_{\text{int}} \left[H'_{\text{int}}(t), \left[H'_{\text{int}}(t'), p_{\pm}(0) \right] \right] \right) = \\ &\quad \hookrightarrow \text{Tr}(A[B, C, D]) = \text{Tr}(D[A[B, C]]) \\ &= - \int_0^t dt' \langle [n_{\text{int}}, H'_{\text{int}}(t)], H'_{\text{int}}(t') \rangle \end{aligned}$$

$$H_e = \sum_j (E_a c_{ja}^\dagger c_{ja} + E_b c_{jb}^\dagger c_{jb}) - \sum_{\langle ij \rangle} \underbrace{(t_a(\vec{r}_j - \vec{r}_i) \cdot c_{ja}^\dagger c_{ia} + t_b(\vec{r}_j - \vec{r}_i) c_{jb}^\dagger c_{ib} + h.c.)}_{\text{da model ne da interband el.-ph int!}} + \sum_{\langle ij \rangle} t_{ab}(\vec{r}_j - \vec{r}_i) c_{aj}^\dagger c_{bi} + h.c.$$

$$\approx 0 + \vec{\nabla}_r t_{ab}(r) \cdot (\vec{u}_j - \vec{u}_i)$$



$$\text{lahko } (a_a, a_b, a_c)$$

$$\vec{d}_{ij,ab} = \vec{r}_{ja} - \vec{r}_{ib} = \underbrace{(\vec{r}_j - \vec{r}_i) + (\vec{r}_a - \vec{r}_b)}_{\vec{d}_{ij,ba}} + (\vec{u}_{ja} - \vec{u}_{ib})$$

$$t_a = t_a(d_{ij,aa}) \text{ in } t_b = t_b(d_{ij,bb})$$

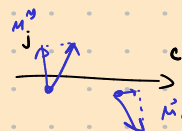
$$\Rightarrow t_a \approx t_a(d_{ij,ss}) + \frac{\partial}{\partial r} (t_a(r)) \bigg|_{d_{ij,ss}} (\vec{u}_{js} - \vec{u}_{is}) \quad R = (u, v)$$

→ assumci: Brskaj odve se atom/orbitale premikajo skupaj: $\vec{u}_{ia} = \vec{u}_{jb} = \vec{u}_i$, etc. (in $\vec{u}_a = \vec{u}_b = 0$)

$$① \quad t_a(d_{ij,aa}) \approx t_a(r_{ij}) + \left(\vec{\nabla}_r t_a(r) \right) \cdot (\vec{u}_j - \vec{u}_i)$$

$$\approx \frac{\partial t_a(r)}{\partial r} \bigg|_{r_{ij}} \hat{R}_{ij} \equiv \alpha_a \hat{R}_{ij} \quad \vec{r} = (x, y)$$

$$② \quad t_{ab}(d_{ij,ab}) \approx 0 + \left(\vec{\nabla}_r t_{ab}(r) \right) \cdot (\vec{u}_j - \vec{u}_i) \approx \alpha_{ab} \cdot \hat{R}_{ij} \cdot (\vec{u}_j - \vec{u}_i)$$



②D

$$\rightarrow \vec{\nabla} [t_{ab}(x, y)] = \vec{\nabla} (xy \cdot f(r)) = (y, x) f(r) + xy \vec{\nabla} f(r) = \underbrace{(y, x) f(r)}_{\phi_a(r) \phi_b(r)} + \underbrace{xy \vec{\nabla} f(r)}_{\approx 0}$$

$$H_e = \sum_i \epsilon_a^{(0)} c_{ia}^\dagger c_{ia} + \epsilon_b^{(0)} c_{ib}^\dagger c_{ib} - \sum_{\langle ij \rangle} (t_{ia} c_{ia}^\dagger c_{ja} + t_{ib} c_{ib}^\dagger c_{jb} + h.c.) + V \sum_j n_j - n_j \quad \text{with } V = \frac{W}{M}$$

$$H_{eg} = \sum_{\langle ij \rangle} \left(d_{ia} c_{ja}^\dagger c_{ia} \hat{R}_{ij}^0 + d_{ib} c_{jb}^\dagger c_{ib} \hat{R}_{ij}^0 + d_{ab} c_{ja}^\dagger c_{ib} + d_{ba} c_{ja}^\dagger c_{ib} + h.c. \right) \cdot (\vec{r}_j - \vec{r}_i)$$

→ first sum is: $V(\vec{r}_i - \vec{r}_j) \approx V \delta(\vec{r}_i - \vec{r}_j) + \mathcal{O}(|\vec{r}_j - \vec{r}_i|)$ & med utveckling se u. Enkelt
coulomblike!
 ↗ no order

$$\vec{r}_j = \frac{1}{\sqrt{N_{im}}} \sum_{\vec{z}, \lambda} \frac{1}{\sqrt{2M\omega_z}} \hat{e}_{\vec{z}, \lambda} e^{i\vec{z} \cdot \vec{r}_j} (b_{\vec{z}, \lambda} + b_{-\vec{z}, \lambda}^\dagger)$$

$$N_{im} = \frac{V}{v_{oc}}$$

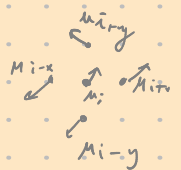
$$c_{aj} = \frac{1}{\sqrt{N_{im}}} \sum_{\vec{r}} e^{i\vec{r} \cdot \vec{r}_j} c_{a\vec{r}}$$

→ new modification: $t_{ab} = 0$. Ampere: $\epsilon_a^{(0)} \rightarrow \epsilon_a^{(0)} + D_a \sum_{\langle ij \rangle} \hat{R}_{ij}^0 (\vec{r}_j - \vec{r}_i) + D_{ab} \sum_{\langle ij \rangle} \sigma_x \hat{R}_{ij}^0 (\vec{r}_j - \vec{r}_i)$

→ 2D model: $\hat{\delta} = (\delta_x, \delta_y) \Rightarrow \sum_{\langle ij \rangle} = \sum_{\vec{r}} \sum_{\delta} (\delta \in \{\pm \vec{x}, \pm \vec{y}\})$

$$\tilde{\delta} = (\delta_x, \delta_y)$$

$$\Rightarrow \vec{r}_j - \vec{r}_i = \sum_{\vec{r}} \sum_{\delta} (\vec{r}_i + \delta - \vec{r}_i)$$



$$\begin{aligned} \sum_{\delta} \hat{\delta} (\vec{r}_{i+\delta} - \vec{r}_i) &= \hat{x} \cdot (\vec{r}_{i+x} - \vec{r}_i) - \hat{x} \cdot (\vec{r}_{i-x} - \vec{r}_i) + \hat{y} \cdot (\vec{r}_{i+y} - \vec{r}_i) - \hat{y} \cdot (\vec{r}_{i-y} - \vec{r}_i) \\ &= (n_{i+x}^x - n_{i-x}^x) + (n_{i+y}^y - n_{i-y}^y) \approx 2\vec{D} \cdot \vec{r} \end{aligned}$$

$$\Rightarrow \sum_{\delta} \tilde{\delta} (\vec{r}_{i+\delta} - \vec{r}_i) = \tilde{y} \cdot (\vec{r}_{i+x} - \vec{r}_i) - \tilde{y} \cdot (\vec{r}_{i-x} - \vec{r}_i) + \dots = (n_{i+x}^y - n_{i-x}^y) - (n_{i+y}^x - n_{i-y}^x)$$

is then defined:

varianella ispeljivna

semo interband

$$H_{e-p} = \sum_{i, \delta} \left(D_a c_{ai}^\dagger c_{ai} + D_b c_{bi}^\dagger c_{bi} + d_a c_{ai}^\dagger c_{i+\delta} + d_b c_{bi}^\dagger c_{i+\delta} \right) \cdot \hat{\delta} (\vec{r}_{i+\delta} - \vec{r}_i) +$$

$$+ \sum_{i, \delta} \left(D_{ab} c_{ai}^\dagger c_{bi} + h.c. \right) \tilde{\delta} \cdot (\vec{r}_{i+\delta} - \vec{r}_i) \quad \leftarrow \text{da interband coupling}$$

$$\vec{r}_{i+\delta} - \vec{r}_i = \frac{1}{N} \sum_{\vec{z}, \lambda} \frac{1}{\sqrt{2M\omega_z}} \hat{e}_{\vec{z}, \lambda} (e^{i\vec{z} \cdot \delta} - 1) e^{i\vec{z} \cdot \vec{r}_i} (b_{\vec{z}, \lambda} + b_{-\vec{z}, \lambda}^\dagger), \quad \hat{e}_{\vec{z}, \lambda} = \vec{e}_{-\vec{z}, \lambda}^*$$



$$\sum_{i \in \vec{\delta}} \delta(\vec{r}_{i,0} - \vec{r}_i) c_{\alpha i}^+ c_{\beta i+s} = \sum_{\vec{z}, \lambda} \frac{(b_{\vec{z}, \lambda} + b_{-\vec{z}, \lambda}^+)}{\sqrt{2NM\omega_{\vec{z}, \lambda}}} \sum_{\vec{\delta}} \hat{\delta} \cdot \vec{\epsilon}_{\vec{z}, \lambda} (e^{i\vec{z}\delta} - 1) \frac{1}{N} \sum_{\vec{r}} e^{i\vec{r}\delta} c_{\alpha \vec{r}}^+ c_{\beta \vec{r}} \sum_i e^{i(\vec{z} + \vec{r} - \vec{r}_i) \cdot \vec{r}_i} =$$

$\delta_{\vec{z} + \vec{r}, \vec{r}} \pmod{\vec{L}}$

$$= \frac{1}{\sqrt{N}} \sum_{\vec{z}, \lambda} \left[\frac{1}{\sqrt{2M\omega_{\vec{z}, \lambda}}} \sum_{\vec{\delta}} \hat{\delta} \cdot \vec{\epsilon}_{\vec{z}, \lambda} (e^{i\vec{z}\delta} - 1) e^{i\vec{r}\delta} \right] c_{\alpha \vec{r} + \vec{z}}^+ c_{\beta \vec{r}} (b_{\vec{z}, \lambda} + b_{-\vec{z}, \lambda}^+)$$

$$\alpha, \beta \in \{1, 2\}$$

$$\vec{z} \cdot (-\vec{x}) = -z_x$$

$$\begin{aligned} \rightarrow \sum_{\delta \in \{\pm x, \pm y\}} (e^{i\vec{z}\delta} - 1) e^{i\vec{r}\delta} \hat{\delta} \cdot \vec{\epsilon}_{\vec{z}} &= \epsilon_{\vec{z}, x}^x \left[(e^{iz_x} - 1) - (e^{-iz_x} - 1) \right] e^{ik_x} \\ &+ \epsilon_{\vec{z}, y}^y \left[(e^{iz_y} - 1) - (e^{-iz_y} - 1) \right] e^{ik_y} \\ &= 2i \left(\sin(k_x + z_x) - \sin(k_x) \right) \end{aligned}$$

$$\Rightarrow H_{ep} = \frac{1}{\sqrt{N}} \sum_{\vec{r}, \vec{z}, \lambda} (c_{\alpha \vec{r} + \vec{z}}^+, c_{\beta \vec{r} + \vec{z}}^+) \begin{pmatrix} g_{aa}^{\dagger}(k, \vec{z}) & g_{ab}^{\dagger} \\ g_{ba}^{\dagger} & g_{bb}^{\dagger} \end{pmatrix} \begin{pmatrix} c_{ak} \\ c_{bk} \end{pmatrix} (b_{\vec{z}, \lambda} + b_{-\vec{z}, \lambda}^+)$$

$$g_{aa}^{\dagger}(k, \vec{z}) = \frac{2i}{\sqrt{2M}} \frac{1}{\sqrt{\omega_{\vec{z}, \lambda}}} \left[\epsilon_{\vec{z}, x}^x \left(D_a \sin(q_x) + \alpha_a \left(\sin(k_x + z_x) - \sin(k_x) \right) \right) + \epsilon_{\vec{z}, y}^y (x \rightarrow y) \right]$$

$\alpha_a \propto \sqrt{\omega_{\vec{z}, \lambda}^2} - \omega(\vec{k})$

$$g_{ab}^{\dagger}(k, \vec{z}) = \frac{2i}{\sqrt{2M}} \frac{1}{\sqrt{\omega_{\vec{z}, \lambda}}} \left[\epsilon_{\vec{z}, x}^x (D_{ab} \sin(q_x)) + \epsilon_{\vec{z}, y}^y (D_{ab} \sin(q_y)) \right]$$

$$H_{ep}^D = \frac{1}{\sqrt{N}} \sum_{\vec{r}, \vec{z}, \lambda} (\alpha_{\vec{r} + \vec{z}}^+, \beta_{\vec{r} + \vec{z}}^+) \underbrace{\begin{pmatrix} \mu_{\vec{r} + \vec{z}} & -\nu_{\vec{r} + \vec{z}} \\ \nu_{\vec{r} + \vec{z}} & \mu_{\vec{r} + \vec{z}} \end{pmatrix}}_{\underline{G}(\vec{k}, \vec{z})} \underline{g}(\vec{k}, \vec{z}) \begin{pmatrix} \mu_k & \nu_k \\ -\nu_k & \mu_k \end{pmatrix} \begin{pmatrix} \alpha_k \\ \beta_k \end{pmatrix} (b_{\vec{z}, \lambda} + b_{-\vec{z}, \lambda}^+)$$

$\rightarrow$ a to symmetric order : large atom separation / near minimum ($D_{ab} \neq 0$)

T.b. el. - form hamiltonke:

$$p(\vec{r}) = -e \psi^\dagger(\vec{r}) \psi(\vec{r})$$

$$\psi(\vec{r}) = \sum_{i\alpha=\{a,b\}} \phi_\alpha(\vec{r}-\vec{r}_i) c_{\alpha i} \quad \bigg| \int d\vec{r} e^{i\vec{z}\cdot\vec{r}}$$

$$p(\vec{z}) = -e \sum_{ij} \sum_{\alpha\beta} \int d\vec{r} e^{-i\vec{z}\cdot\vec{r}} \phi_\alpha^\dagger(\vec{r}-\vec{r}_i) \phi_\beta(\vec{r}-\vec{r}_j) = -e \sum_{ij} \sum_{\alpha\beta} e^{i\vec{z}\cdot\vec{r}_i} \langle \phi_\alpha | e^{i\vec{z}\cdot\vec{x}} | \phi_\beta \rangle c_{\alpha i}^\dagger c_{\beta j} \delta_{ij}$$

$\vec{r} = \vec{r}_i + \vec{x}$

$= 1 - i\vec{z}\cdot\vec{x}$
↑
dipol

$$\rightarrow p(\vec{z}) \approx -e \sum_i e^{-i\vec{z}\cdot\vec{r}_i} \left(n_{ai} + n_{bi} - i(\vec{z} \cdot \overbrace{\langle \phi_a | \vec{x} | \phi_b \rangle}^{\vec{d}}) \cdot (c_{ai}^\dagger c_{bi} + c_{bi}^\dagger c_{ai}) \right)$$

$$\rightarrow \sum_i e^{i\vec{z}\cdot\vec{r}_i} c_{ai}^\dagger c_{bi} = \frac{1}{N} \sum_k c_{a, k+\vec{z}}^\dagger c_{b, k}$$

$\frac{1}{\sqrt{N}} \sum_k c_{a, k}^\dagger e^{-ik\cdot\vec{r}_i}$

assume samo a, b dipol.

$$p(-\vec{z}) = -e \sum_k (c_{a, k+\vec{z}}^\dagger, c_{b, k+\vec{z}}^\dagger) \left(1 + i\vec{z} \cdot \overbrace{\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}}^{\vec{d}_x} \right) \begin{pmatrix} c_{a, k} \\ c_{b, k} \end{pmatrix}$$

$$\rightarrow H_{ex} = - \int d\vec{r} \vec{j}_e \cdot \vec{A} = - \int d\vec{r} \sum_{\vec{z}} \vec{A}_{\vec{z}} e^{i\vec{z}\cdot\vec{r}} \cdot \vec{j}_e(\vec{r}) = - \sum_{\vec{z}} \vec{A}_{\vec{z}} \cdot \left(\int d\vec{r} e^{i\vec{z}\cdot\vec{r}} \vec{j}_e(\vec{r}) \right) = - \sum_{\vec{z}} \vec{j}_e(\vec{z}) \cdot \vec{A}(\vec{z})$$

$\vec{j}_e(\vec{z})$ (±)?

$$\vec{j}_e(\vec{r}) = -\frac{e}{im} \left(\psi^\dagger(\vec{r}) (-i\vec{\nabla}) \psi(\vec{r}) \right) \rightarrow \text{to } j \text{ to } \text{for el.}$$

→ toh r TB model:

Heuristik: $\frac{\partial p(\vec{r})}{\partial t} = i [H_{e, f}] \quad \left(\dot{m} \right) \quad \frac{\partial p}{\partial t} + \vec{\nabla} \cdot \vec{j} = 0$

$$H_e = \sum_i \epsilon_a^{(i)} c_{ai}^\dagger c_{ai} + \epsilon_b^{(i)} c_{bi}^\dagger c_{bi} - \sum_{(ij)} (t_{aia} c_{ia}^\dagger c_{ja} + t_{bia} c_{bi}^\dagger c_{ja} + h.c.) \left(+ V \sum_j n_j \right)$$

$$i\vec{z} \cdot \vec{j}(\vec{z}) = i \left[H_{e, f}(\vec{z}) \right]$$

$$H'_e = \sum_{\vec{r}=\{a,b\}} \epsilon_{\vec{r}, k} c_{\vec{r}, k}^\dagger c_{\vec{r}, k}$$

$$-e \sum_{k, k'} \left[\psi_k^\dagger \begin{pmatrix} \epsilon_{ak} & 0 \\ 0 & \epsilon_{bk} \end{pmatrix} \psi_{k'} + \psi_{k-\vec{z}}^\dagger \begin{pmatrix} 1 & -i\vec{z}\cdot\vec{d} \\ -i\vec{z}\cdot\vec{d} & 1 \end{pmatrix} \psi_{k'} \right] =$$

→ identitiki: $[\psi_k^\dagger A \psi_k, \psi_{k'-\vec{z}}^\dagger B \psi_{k'}] = \psi_{k'-\vec{z}}^\dagger (B A(k') - A(k'-\vec{z}) B) \psi_{k'} \delta_{k, k'-\vec{z}}$

$= \psi_k^\dagger (B A(k+\vec{z}) - A(k) B) \psi_{k+\vec{z}}$

$$= -e \sum_{\vec{k}} \psi_{\vec{k}}^{\dagger} \left[\begin{pmatrix} \epsilon_{a, \vec{k}+\vec{q}} & -i(\vec{q} \cdot \vec{d}) \cdot \epsilon_{b, \vec{k}+\vec{q}} \\ -i(\vec{q} \cdot \vec{d}) \cdot \epsilon_{a, \vec{k}} & \epsilon_{b, \vec{k}+\vec{q}} \end{pmatrix} - \begin{pmatrix} \epsilon_{a, \vec{k}} & -i(\vec{q} \cdot \vec{d}) \cdot \epsilon_{b, \vec{k}} \\ -i(\vec{q} \cdot \vec{d}) \cdot \epsilon_{a, \vec{k}} & \epsilon_{b, \vec{k}} \end{pmatrix} \right] \psi_{\vec{k}+\vec{q}}$$

→ here for $\vec{q} \approx 0 \rightarrow \epsilon_{a, \vec{k}+\vec{q}} \sim \epsilon_{a, \vec{k}} + \vec{q} \cdot \nabla_{\vec{k}} \epsilon_{a, \vec{k}} \quad (*)$

→ $\epsilon_{a, \vec{k}+\vec{q}} - \epsilon_{a, \vec{k}} \approx \vec{q} \cdot \nabla_{\vec{k}} \epsilon_{a, \vec{k}}$

→ $\vec{q} \cdot \vec{j}(\vec{q}) \approx \vec{q} \cdot \left(-e \sum_{\vec{k}} \psi_{\vec{k}}^{\dagger} \begin{pmatrix} \nabla_{\vec{k}} \epsilon_{a, \vec{k}+\vec{q}} & -i(\epsilon_{b, \vec{k}+\vec{q}} - \epsilon_{a, \vec{k}}) \cdot \vec{d} \\ -i\vec{d}(\epsilon_{a, \vec{k}+\vec{q}} - \epsilon_{b, \vec{k}}) & \nabla_{\vec{k}} \epsilon_{b, \vec{k}+\vec{q}} \end{pmatrix} \psi_{\vec{k}} \right) \Rightarrow$

→ $\vec{j}(\vec{q}) = -e \sum_{\vec{k}} \psi_{\vec{k}-\vec{q}}^{\dagger} \begin{pmatrix} \nabla_{\vec{k}} \epsilon_{a, \vec{k}-\vec{q}} & +i\vec{d}(\epsilon_{a, \vec{k}-\vec{q}} - \epsilon_{b, \vec{k}}) \\ +i\vec{d}(\epsilon_{b, \vec{k}-\vec{q}} - \epsilon_{a, \vec{k}}) & \nabla_{\vec{k}} \epsilon_{b, \vec{k}-\vec{q}} \end{pmatrix} \psi_{\vec{k}}$

→ $\left(\begin{array}{l} \text{zakaj } \odot \\ \text{primer za free el.} \rightarrow \end{array} \frac{(\hbar v)^2}{2m} - \frac{\hbar^2}{2m} = \frac{1}{2m} (2\vec{q} \cdot \vec{k} + q^2) = \vec{q} \cdot \frac{1}{2m} (2\vec{k} + \vec{q}) = \vec{q} \cdot \left(\frac{\vec{k} + \frac{\vec{q}}{2}}{m} \right) \right)$

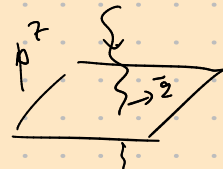
+ b. → $-\cos(k_q) + \cos(k) = 2 \sin \frac{q}{2} \sin \left(k + \frac{q}{2} \right) \approx 2 \cdot \frac{\sin \left(k + \frac{q}{2} \right)}{\nabla_{\vec{k}} \epsilon_{k+\frac{q}{2}}}$

$H_{eg} = -\sum_{\vec{q}} \vec{j}(\vec{q}) \cdot \vec{A}(\vec{q}) = \sum_{\vec{q}} \frac{e}{\sqrt{2V\epsilon_0\omega_{\vec{q}}}} \psi_{\vec{k}-\vec{q}}^{\dagger} \left(\begin{pmatrix} \vec{\epsilon}_{\vec{q}} \cdot \vec{p}_a(\vec{k} + \frac{\vec{q}}{2}) & +i(\vec{\epsilon}_{\vec{q}} \cdot \vec{d})(\epsilon_{a, \vec{k}+\vec{q}} - \epsilon_{b, \vec{k}}) \\ +i(\vec{\epsilon}_{\vec{q}} \cdot \vec{d})(\epsilon_{b, \vec{k}+\vec{q}} - \epsilon_{a, \vec{k}}) & \vec{\epsilon}_{\vec{q}} \cdot \vec{p}_b(\vec{k} + \frac{\vec{q}}{2}) \end{pmatrix} \right) \psi_{\vec{k}} (a_{\vec{q}, \lambda} + a_{-\vec{q}, \lambda}^{\dagger})$

→ $M^A(\vec{k}, \vec{q})$

$k - q \rightarrow k + q$ (nekeje je narobe, preveri kje si jati minus.)

→ simple polarizacije : setup:



$\vec{q} = q(\cos \varphi, \sin \varphi)$

$\vec{E}_{TE}(z) = (\hat{z} \times \vec{q}) \sin\left(\frac{\pi}{L} z\right)$

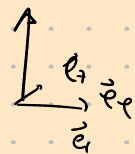
$\vec{E}_{TM}(z) = \frac{\omega_0}{\omega_z} \hat{z} \sin\left(\frac{\pi}{L} z\right) + i \frac{c z}{\omega_f} \hat{z} \cos\left(\frac{\pi}{L} z\right) \rightarrow z = \frac{L}{2} \Rightarrow$

$\vec{E}_{TE} \approx (-\sin \varphi, \cos \varphi)$
 $\vec{E}_{TM} \approx \frac{\omega_0}{\omega_z} (\cos \varphi, \sin \varphi)$

→ $\vec{p}_{a,b}(\vec{k} + \frac{\vec{q}}{2}) = 2\hbar a_{\vec{k}} \left(\sin\left(k_x + \frac{q_x}{2}\right), \sin\left(k_y + \frac{q_y}{2}\right) \right), \quad \vec{d} = \begin{cases} d(\cos \theta, \sin \theta) & \text{in plane} \\ d\hat{z} & \text{out of plane} \end{cases}$

	TE	TM
$\vec{E} \cdot \vec{r}_i (k_x \frac{L}{2})$	$2 \cos \left(-\sin \gamma \sin \left(k_x + \frac{2x}{L} \right) + \cos \gamma \sin \left(k_y + \frac{2y}{L} \right) \right)$	$\frac{\omega_0}{\omega_i} 2 \cos \left(\cos \gamma \sin \left(k_x + \frac{2x}{L} \right) + \sin \gamma \sin \left(k_y + \frac{2y}{L} \right) \right)$
$\vec{E} \cdot \vec{d}$ in plane	$d \sin(\phi - \varphi)$ is k and q dependent.	$\frac{\omega_0}{\omega_i} d \cos(\phi - \varphi)$
$\vec{E} \cdot \vec{d}$ canc axis	0	0 in $z = \frac{L}{2}$

$$\vec{d} = \langle \phi_a | \vec{r} | \phi_b \rangle = \int \phi_a^*(\vec{r} - \vec{r}_i) \vec{r} \phi_b(\vec{r} - \vec{r}_i) d\vec{r} \quad / \cdot \hat{e}_x$$

$$d_y \stackrel{①}{=} d_x = \int \phi_a^*(\vec{r} - \vec{r}_i) r \cos \theta \phi_b(\vec{r} - \vec{r}_i) d\vec{r} \dots$$


$$\rightarrow \text{relevant to } \sum_{\lambda} |\vec{C}_{\lambda} \cdot \vec{X}|^2 = |\hat{z} \times \vec{X}|^2 + \frac{\omega_i^2}{\omega_0^2} |\hat{z} \cdot \vec{X}|^2 \xrightarrow{\vec{z} \rightarrow 0} |\vec{X}|^2$$

$$\rightarrow \text{vector: } 4 \cos^2 \left(\sin^2 k_x + \sin^2 k_y \right), \text{ in order of real axis coordinates!}$$

Electron-photon coupling zu free el.

$$\psi(\vec{r}) = \frac{1}{\sqrt{V}} \sum_{\vec{k}} e^{i\vec{k} \cdot \vec{r}} c_{\vec{k}}$$

$$H_e = \sum_{\vec{k}} \underbrace{(\epsilon_0 + \hbar k^2)}_{\epsilon_{\vec{k},0}} + H_v$$

$$= \sum_{\vec{k} \in \{a, b\}} \epsilon_{\vec{k},0} + H_v$$

$$\pm \frac{G}{2} \mp 2t(\cos k_x + \cos k_y)z$$

$$k_x \sim k_y \sim 0$$

$$= \pm \frac{G}{2} \mp 2t \left(1 - \frac{k_x^2}{2} + 1 - \frac{k_y^2}{2} \right) =$$

$$= \left(\pm \frac{G}{2} \mp 4t \right) \pm t(k_x^2 + k_y^2) = \pm \left(\epsilon^0 + \hbar \vec{k} \cdot \vec{k} \right)$$

$$\epsilon^0 = \frac{G}{2} - 4t$$

$$H_{ey}^{(1)} = - \int \vec{j}_e \cdot \vec{A} d\vec{r} = - \sum_{\vec{k}} \vec{j}_e(-\frac{\vec{k}}{2}) A(\frac{\vec{k}}{2})$$

$$\vec{j}_e(\vec{r}) = -\frac{e}{2m} (\psi^\dagger(\vec{r}) \vec{\nabla} \psi(\vec{r}) - (\vec{\nabla} \psi^\dagger(\vec{r})) \psi(\vec{r})) = -\frac{e}{2m} \frac{1}{V} \sum_{\vec{k}, \vec{k}'} \left(e^{-i\vec{k}' \cdot \vec{r}} e^{i\vec{k} \cdot \vec{r}} i\vec{k} c_{\vec{k}'}^\dagger c_{\vec{k}} - e^{i\vec{k}' \cdot \vec{r}} e^{-i\vec{k} \cdot \vec{r}} (-i\vec{k}') c_{\vec{k}'}^\dagger c_{\vec{k}} \right)$$

$$\vec{j}_e(-\frac{\vec{k}}{2}) = \int d\vec{r} e^{i\frac{\vec{k}}{2} \cdot \vec{r}} \vec{j}_e(\vec{r}) = -\frac{e}{2m} \sum_{\vec{k}, \vec{k}'} \underbrace{\left(\frac{1}{V} \int d\vec{r} e^{i(\frac{\vec{k}}{2} + \vec{k} - \vec{k}') \cdot \vec{r}} \right)}_{\delta_{\vec{k}', \vec{k} + \frac{\vec{k}}{2}}} c_{\vec{k}'}^\dagger c_{\vec{k}} \vec{k} + \underbrace{\left(\frac{1}{V} \int d\vec{r} e^{i(\vec{k} + \vec{k} - \vec{k}') \cdot \vec{r}} \right)}_{\delta_{\vec{k}', \vec{k} + \vec{k}}} c_{\vec{k}'}^\dagger c_{\vec{k}} \vec{k} =$$

$$= -\frac{e}{2m} \sum_{\vec{k}} \left(c_{\vec{k} + \frac{\vec{k}}{2}}^\dagger c_{\vec{k}} \vec{k} + (\vec{k} + \frac{\vec{k}}{2}) c_{\vec{k} + \frac{\vec{k}}{2}}^\dagger c_{\vec{k}} \right) =$$

$$= -\frac{e}{m} \sum_{\vec{k}} \left(\vec{k} + \frac{\vec{k}}{2} \right) \cdot \psi_{\vec{k} + \frac{\vec{k}}{2}}^\dagger \psi_{\vec{k}}$$

$$\rightarrow H_{ey}^{(1)} = \frac{e}{m} \sum_{\vec{k} \in \{a\}} \left(\vec{k} + \frac{\vec{k}}{2} \right) \cdot \psi_{\vec{k} + \frac{\vec{k}}{2}}^\dagger \psi_{\vec{k}} (a_{\frac{\vec{k}}{2},1} + a_{-\frac{\vec{k}}{2},1}^\dagger) \frac{1}{\sqrt{2V\epsilon_0 \omega_{\vec{k},1}}}$$

El. -phona coupling in free el.:

$$\varepsilon_i^0 = \pm \left(\frac{\hbar}{2} - 4t_3 \right)$$

$$H_e = \sum_{\vec{k}_0} \left(\varepsilon_0^0 + t_8 k^2 \right) + H_v$$

$\downarrow$

(?) model:

